

Year 7 · Chapter 6 · Geometrical figures: answers (teacher copy)

6.01 Transforming shapes (booklet p. 3)

WE DO examples (teacher)

- Column 1, E1: (5, 2): 2 right, 4 down → (7, -2)
- Column 1, E2: (-2, 3): 6 left, 3 down → (-8, 0)
- Column 2, E1: (-8, 5) in the x-axis → change the sign of y; → (-8, -5)
- Column 2, E2: (-8, -6) in the x-axis → change the sign of y; → (-8, 6)
- Column 3, E1: Rotate (2, -2) by 180° about O. → 180°: (x, y) → (-x, -y); (2, -2) → (-2, 2)
- Column 3, E2: Rotate (3, -4) 90° clockwise about O. → 90° clockwise: (x, y) → (y, -x); (3, -4) → (-4, -3)
- Column 3, E3: Rotate (-4, 6) 90° anticlockwise about O. → 90° anticlockwise: (x, y) → (-y, x); (-4, 6) → (-6, -4)

Questions (in order: Easy, Medium, Challenging)

- | | | |
|-------------|--------------|--------------|
| 1. (0, 2) | 13. (12, 1) | 25. (5, 6) |
| 2. (7, 11) | 14. (8, 0) | 26. (-7, -7) |
| 3. (-5, 2) | 15. (-3, 0) | 27. (-6, -2) |
| 4. (-9, -9) | 16. (6, -4) | 28. (6, 8) |
| 5. (5, -5) | 17. (-9, -1) | 29. (5, 6) |
| 6. (0, 2) | 18. (-2, 5) | 30. (-4, 2) |
| 7. (4, 3) | 19. (8, -5) | 31. (-6, 5) |
| 8. (-6, 6) | 20. (2, 3) | 32. (6, -3) |
| 9. (-7, 10) | 21. (-8, 5) | 33. (6, 2) |
| 10. (5, -4) | 22. (5, 3) | 34. (-6, -5) |
| 11. (4, -9) | 23. (-6, -7) | 35. (-3, -1) |
| 12. (2, -3) | 24. (-3, 5) | |

6.02 Combined transformations (booklet p. 4)

WE DO examples (teacher)

- Column 1, E1: a slide → translation
- Column 1, E2: a flip → reflection
- Column 1, E3: (diagram) → translation
- Column 2, E1: 7 right, 6 up; then 6 left, 2 down → across: 7 - 6 = 1; up/down: 6 - 2 = 4; 1 right, 4 up
- Column 2, E2: 1 right, 6 up; then 6 left, 2 down → across: 1 - 6 = -5; up/down: 6 - 2 = 4; 5 left, 4 up
- Column 3, E1: (-2, -3): 4 right, then reflect in the x-axis → 4 right: (2, -3); reflect in x-axis: (2, 3)
- Column 3, E2: (4, 2): rotate 180° about O, then 5 left → rotate 180°: (-4, -2); 5 left: (-9, -2)

Questions (in order: Easy, Medium, Challenging)

- | | | |
|----------------|---------------------|--------------|
| 1. rotation | 10. reflection | 19. (-3, -4) |
| 2. reflection | 11. reflection | 20. (-2, -4) |
| 3. translation | 12. rotation (180°) | 21. (4, 4) |
| 4. rotation | 13. translation | 22. (-6, -1) |
| 5. reflection | 14. 4 left, 2 up | 23. (0, -3) |
| 6. translation | 15. 3 right, 3 up | 24. (2, -3) |
| 7. rotation | 16. 1 right, 1 down | 25. (1, -3) |
| 8. rotation | 17. 2 right, 1 up | 26. (0, 3) |
| 9. translation | 18. 1 left, 3 down | |

6.03 Line symmetry (booklet p. 5)

WE DO examples (teacher)

- Column 1, E1: (diagram) → 4
- Column 1, E2: (diagram) → 2
- Column 2, E1: a regular polygon with 16 sides → 16 sides → 16 axes; answer: 16
- Column 2, E2: a regular polygon with 3 sides → 3 sides → 3 axes; answer: 3

- Column 3, E1: Name it and draw the axes. → shape: square; axes of symmetry: 4
- Column 3, E2: Name it and draw the axes. → shape: rectangle; axes of symmetry: 2

Questions (in order: Easy, Medium, Challenging)

- | | | |
|------|--------|----------------|
| 1. 2 | 9. 5 | 17. 4 |
| 2. 0 | 10. 13 | 18. 18 |
| 3. 1 | 11. 7 | 19. 9 |
| 4. 1 | 12. 12 | 20. rhombus: 2 |
| 5. 3 | 13. 20 | 21. kite: 1 |
| 6. 1 | 14. 19 | 22. isosceles |
| 7. 0 | 15. 15 | trapezium: 1 |
| 8. 6 | 16. 10 | |

6.04 Rotational symmetry (booklet p. 6)

WE DO examples (teacher)

- Column 1, E1: (diagram) → 4
- Column 1, E2: (diagram) → 2
- Column 2, E1: 24 sides → order 24; $360^\circ \div 24 = 15^\circ$
- Column 2, E2: 15 sides → order 15; $360^\circ \div 15 = 24^\circ$
- Column 3, E1: a square → axes of symmetry: 4; order of rotation: 4
- Column 3, E2: a rectangle → axes of symmetry: 2; order of rotation: 2

Questions (in order: Easy, Medium, Challenging)

- | | | |
|-------------------------|--------------------------|--------------------------|
| 1. 2 | 11. order 36, 10° | 21. order 12, 30° |
| 2. 2 | 12. order 10, 36° | 22. 2 axes; order 2 |
| 3. 1 | 13. order 30, 12° | 23. 0 axes; order 2 |
| 4. 1 | 14. order 18, 20° | 24. 1 axes; order 1 |
| 5. 3 | 15. order 3, 120° | 25. 1 axes; order 1 |
| 6. 1 | 16. order 8, 45° | 26. 3 axes; order 3 |
| 7. 1 | 17. order 20, 18° | 27. 1 axes; order 1 |
| 8. 6 | 18. order 6, 60° | 28. 0 axes; order 1 |
| 9. 5 | 19. order 4, 90° | 29. 6 axes; order 6 |
| 10. order 5, 72° | 20. order 9, 40° | 30. 5 axes; order 5 |

6.05 Classifying triangles (booklet p. 7)

WE DO examples (teacher)

- Column 1, E1: $40^\circ, 85^\circ, 55^\circ$ → acute
- Column 1, E2: $50^\circ, 90^\circ, 40^\circ$ → right
- Column 1, E3: 3 cm, 3 cm, 3 cm → equilateral
- Column 2, E1: (diagram) → angles: right; sides: isosceles
- Column 2, E2: (diagram) → angles: right; sides: scalene
- Column 3, E1: a right-angled isosceles triangle → yes; because: $90^\circ, 45^\circ, 45^\circ$
- Column 3, E2: a triangle with two right angles → no; because: $90^\circ + 90^\circ = 180^\circ$ leaves 0°

Questions (in order: Easy, Medium, Challenging)

- | | | |
|-------------|---------------------|--|
| 1. right | 14. isosceles | 26. isosceles, obtuse |
| 2. acute | 15. isosceles | |
| 3. right | 16. equilateral | 27. scalene, right |
| 4. acute | 17. isosceles | 28. no: equilateral angles are all 60° |
| 5. acute | 18. equilateral | |
| 6. acute | 19. scalene | 29. yes: e.g. $90^\circ, 60^\circ, 30^\circ$ |
| 7. right | 20. scalene | |
| 8. obtuse | 21. equilateral | 30. no: two angles over 90° add to over 180° |
| 9. obtuse | 22. isosceles | |
| 10. acute | 23. scalene | |
| 11. obtuse | 24. equilateral | 31. yes: e.g. $120^\circ, 30^\circ, 30^\circ$ |
| 12. obtuse | 25. scalene, obtuse | |
| 13. scalene | | |

32. yes: e.g. 50° , 60° , 70°
 33. no: equilateral angles are all 60°

6.06 Angle sum of a triangle (booklet p. 8)

WE DO examples (teacher)

- Column 1, E1: 31° and $32^\circ \rightarrow 117^\circ$
- Column 1, E2: 101° and $44^\circ \rightarrow 35^\circ$
- Column 2, E1: Find a . $\rightarrow a + 30 + 30 = 180$; $a = 180 - 60 = 120$
- Column 2, E2: Find a . $\rightarrow a + 70 + 30 = 180$; $a = 180 - 100 = 80$
- Column 3, E1: The apex angle is 24° . Find the base angles x . $\rightarrow 2x + 24 = 180$; $2x = 156$; $x = 78$
- Column 3, E2: A base angle is 80° . Find the apex angle y . $\rightarrow y + 80 + 80 = 180$; $y = 180 - 160$; $y = 20$

Questions (in order: Easy, Medium, Challenging)

- | | | |
|----------------|----------------|-----------------|
| 1. 98° | 14. 79° | 27. 102° |
| 2. 60° | 15. 33° | 28. 40° |
| 3. 98° | 16. 23° | 29. 31° |
| 4. 43° | 17. 99° | 30. 41° |
| 5. 122° | 18. 64° | 31. $a = 75$ |
| 6. 39° | 19. 94° | 32. $a = 80$ |
| 7. 116° | 20. 60° | 33. $a = 75$ |
| 8. 25° | 21. 39° | 34. $x = 64$ |
| 9. 77° | 22. 26° | 35. $y = 70$ |
| 10. 70° | 23. 78° | 36. $x = 38$ |
| 11. 30° | 24. 41° | 37. $y = 40$ |
| 12. 32° | 25. 65° | 38. $x = 32$ |
| 13. 46° | 26. 75° | 39. $y = 50$ |

6.07 Exterior angle of a triangle (booklet p. 9)

WE DO examples (teacher)

- Column 1, E1: interior opposite: 35° and $54^\circ \rightarrow 89^\circ$
- Column 1, E2: interior opposite: 90° and $19^\circ \rightarrow 109^\circ$
- Column 2, E1: The interior opposite angles are 45° and 65° . $\rightarrow x = 45 + 65$ (exterior angle); $x = 110$
- Column 2, E2: The interior opposite angles are 60° and 35° . $\rightarrow x = 60 + 35$ (exterior angle); $x = 95$
- Column 3, E1: The exterior angle is 160° and one interior opposite angle is 65° . Find the other, y . $\rightarrow y + 65 = 160$; $y = 160 - 65$; $y = 95$
- Column 3, E2: The exterior angle is 140° and one interior opposite angle is 95° . Find the other, y . $\rightarrow y + 95 = 140$; $y = 140 - 95$; $y = 45$

Questions (in order: Easy, Medium, Challenging)

- | | | |
|----------------|-----------------|---------------|
| 1. 142° | 10. 147° | 19. $x = 105$ |
| 2. 50° | 11. 93° | 20. $x = 120$ |
| 3. 114° | 12. 105° | 21. $x = 100$ |
| 4. 145° | 13. 145° | 22. $y = 40$ |
| 5. 100° | 14. 103° | 23. $y = 35$ |
| 6. 119° | 15. 82° | 24. $y = 80$ |
| 7. 150° | 16. 76° | 25. $y = 25$ |
| 8. 85° | 17. 90° | 26. $y = 35$ |
| 9. 105° | 18. 128° | |

6.08 Quadrilaterals (booklet p. 10)

WE DO examples (teacher)

- Column 1, E1: 4 equal sides and 4 right angles $\rightarrow$ square
- Column 1, E2: 4 right angles, opposite sides equal $\rightarrow$ rectangle
- Column 2, E1: (diagram) $\rightarrow$ name: square; property: all sides equal and all angles 90°

- Column 2, E2: (diagram) $\rightarrow$ name: rectangle; property: opposite sides equal, all angles 90°
- Column 3, E1: Every square is a rectangle. $\rightarrow$ true; because it has 4 right angles and opposite sides equal
- Column 3, E2: Every rectangle is a square. $\rightarrow$ false; because its sides need not all be equal

Questions (in order: Easy, Medium, Challenging)

- | | | |
|--|---|---|
| 1. rhombus | 9. parallelogram: opposite sides parallel and equal | 12. true: both pairs of opposite sides are parallel |
| 2. parallelogram | | |
| 3. trapezium | | |
| 4. kite | | |
| 5. square | 10. trapezium: exactly one pair of parallel sides | 13. false: its opposite sides are not parallel |
| 6. rectangle | | |
| 7. rhombus | | |
| 8. rhombus: all sides equal, opposite angles equal | 11. kite: two pairs of equal adjacent sides | 14. true: all four sides are equal |
| | | 15. false: it has exactly one pair |

6.09 Angle sum of a quadrilateral (booklet p. 11)

11)

WE DO examples (teacher)

- Column 1, E1: 60° , 115° , $105^\circ \rightarrow 80^\circ$
- Column 1, E2: 90° , 115° , $115^\circ \rightarrow 40^\circ$
- Column 2, E1: Find a . (Not to scale.) $\rightarrow a + 110 + 75 + 90 = 360$; $a = 360 - 275 = 85$
- Column 2, E2: Find a . (Not to scale.) $\rightarrow a + 85 + 90 + 110 = 360$; $a = 360 - 285 = 75$
- Column 3, E1: A parallelogram has an angle of 100° . Find the other three angles. $\rightarrow$ opposite angle: 100° ; co-interior: $180 - 100 = 80^\circ$; angles: 100° , 80° , 100° , 80°
- Column 3, E2: A kite has two equal angles x° , and its other angles are 100° and 40° . Find x . $\rightarrow 2x + 100 + 40 = 360$; $2x = 220$; $x = 110$

Questions (in order: Easy, Medium, Challenging)

- | | | |
|-----------------|-----------------|--|
| 1. 40° | 13. 155° | 25. 70° |
| 2. 75° | 14. 95° | 26. 55° |
| 3. 90° | 15. 105° | 27. 65° |
| 4. 85° | 16. 50° | 28. 90° |
| 5. 115° | 17. 105° | 29. 125° |
| 6. 55° | 18. 115° | 30. $a = 105$ |
| 7. 40° | 19. 140° | 31. $a = 80$ |
| 8. 45° | 20. 35° | 32. $a = 75$ |
| 9. 75° | 21. 40° | 33. 30° , 150° , 30° |
| 10. 90° | 22. 40° | 34. $x = 90$ |
| 11. 155° | 23. 90° | 35. 130° , 50° , 130° |
| 12. 40° | 24. 55° | 36. $x = 80$ |

6.10 Properties of quadrilaterals (booklet p. 12)

12)

WE DO examples (teacher)

- Column 1, E1: The diagonals of a rectangle are equal. $\rightarrow$ true
- Column 1, E2: The diagonals of a rhombus meet at 90° . $\rightarrow$ true
- Column 2, E1: A rectangle has one diagonal 5 cm. How long is the other? $\rightarrow$ rectangle: diagonals are equal; 5 cm
- Column 2, E2: A rhombus has a side of 9 cm. Find its perimeter. $\rightarrow$ rhombus: all sides equal; $4 \times 9 = 36$ cm
- Column 3, E1: A square is a rectangle. $\rightarrow$ because it has four right angles; and opposite sides equal
- Column 3, E2: A square is a rhombus. $\rightarrow$ because it has four equal sides; and opposite sides parallel

Questions (in order: Easy, Medium, Challenging)

1. false
2. true
3. true
4. false
5. true

6. false
7. true
8. true
9. false
10. true

11. 65°
12. 8 cm
13. 40°
14. 7 cm
15. 24 cm

16. 60°

17. 12 cm

18. 145°

19. both pairs of
opposite sides
are parallel

20. both pairs of
opposite sides
are parallel
and equal

21. it has two pairs
of equal
adjacent sides

22. it has two pairs
of equal
adjacent sides